Magnificent Spiral Speakers Shaped Like Nautilus

We come across speakers in various shapes and sizes in our daily life. These are made to give the best performance possible within the size and price constraints. But what if we did not have any constraint, and wanted to produce the best possible speakers? Nature has solution to most problems, including this. Read on to find out, how

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It appears that Mother Nature has a general tendency to create objects in the shape of logarithmic spirals like arms of spiral galaxies, human cochlea, nautilus—a type of molluscs found in deep sea, spirula—a deep water squid, etc. The galaxies in space are differentially rotating, which causes a disturbance in the disc to wind up into a spiral form. The stars and gas in the disk of a galaxy exert considerable gravitational force that helps maintain the spiral form against the tendency to disperse. The tight wound spiral shape of cochlea—in contrast to the stretched-out version found in birds or reptiles—is useful for packing a slew of hearing parts into a very small space. Another benefit is that the distinctive snail shape enhances the ability of the ear to detect low-frequency sound.

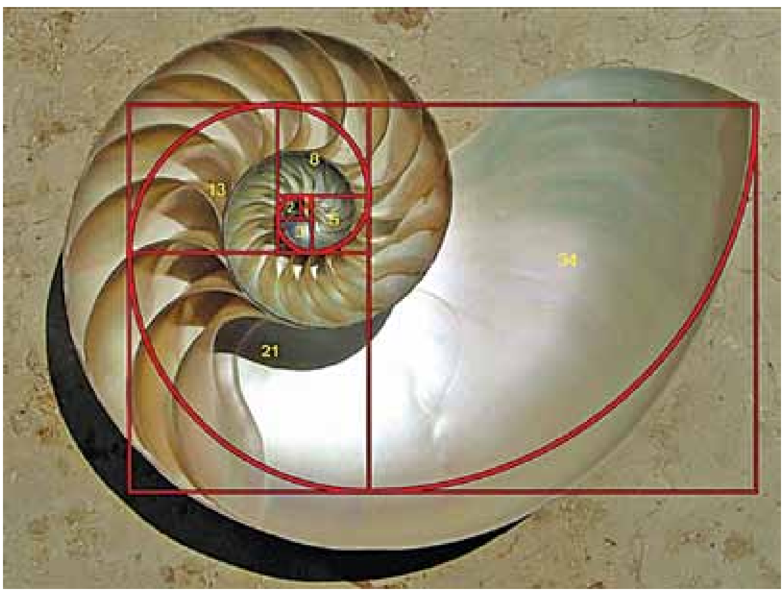

Fig. 1: Logarithmic geometry of a nautilus shell

Fibonacci was one of the most famous mathematicians who in 1202 introduced the Fibonacci sequence 0, 1, 1, 2, 3, 5, 8, 13, 21, and so on. As you may have noticed, each number in Fibonacci sequence is the sum of previous two numbers. By drawing circular arcs connecting the opposite corners of squares, whose side lengths are successive Fibonacci numbers, that is, 1, 1, 2, 3, 5, 8, 13, 21, and 34, a logarithmic spiral can be created.

Nautilus mollusc

The nautilus is a marine mollusc found in Indo-Pacific at a depth of several hundred metres. The nautilus shell presents one of the finest natural examples of a logarithmic spiral as can be seen in Fig. 1.

A section cut of a nautilus shell is shown in Fig. 2. Its logarithmic coiled tapered tube saves space, and when it grows, it need not change its shell. Inspired by the natural geometry of the shell of nautilus and understanding its function, audio engineers adopted logarithmic spiral shape to construct a sound system that not only looks great but also sounds better.

High-fidelity sound reproduction

Speakers require an enclosure to reproduce sound faithfully. The enclosure prevents the sound waves generated by the rear-facing diaphragm of an open speaker driver from interacting with the sound waves generated at front of the speaker driver. Because the forward and rearward generated sounds are out of phase with each other, any interaction between the two in the listening space creates a distortion of the original signal that was